NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-94/7A
12 October 1989
SUPERSEDING
~~MIL-R-94/7~~
~~21 May 1986~~

PERFORMANCE SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND, PRINTED CIRCUIT BOARD MOUNTING, STYLE RV8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RV8, printed circuit board mounting, nonwire-wound, variable, resistors.

2. APPLICABLE DOCUMENT

2.1 Government document.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-94 - Resistors, Variable, Composition, General Specification for.

STANDARD

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-50099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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FSC 5905

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3. REQUIREMENTS

3.1 Requirements. The individual item requirements shall be in accordance with MIL-R-94, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Mounting and locking nuts. The mounting nut shall be .078 inch (1.98 mm) thick and shall measure .312 inch (7.92 mm) across the hexagonal flats. For locking bushings, the locking nut shall be .156 inch (3.86 mm) thick and shall measure .312 inch (7.92 mm) across the hexagonal flats. Thread size shall be .250-32UNEF-2B.

3.2.2 Internal-tooth lockwasher. An internal-tooth lockwasher shall be supplied and when mounted, shall have a maximum thickness of approximately 0.045 inch (1.14 mm).

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.032 inch (0.81 mm).

3.3 Total mechanical rotation. The total mechanical rotation without switch shall be within the limits of 295° ±3°.

3.4 Shaft length. The length of the operating shaft shall be in accordance with table I.

TABLE I. Length of operating shaft. 1/

Symbol	Length of operating shaft from mounting surface of resistor (nominal)		
	Flatted	Slotted	
	Bushings N and S (inch, ±.03125)	Bushings N and S (inch, ±.03125)	Bushings L and T (inch, ±.03125)
A	---	5/8	5/8
B	---	1/2	---
D	7/8	7/8	7/8
E	---	---	---
G	---	---	---
L	---	3/8	---

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

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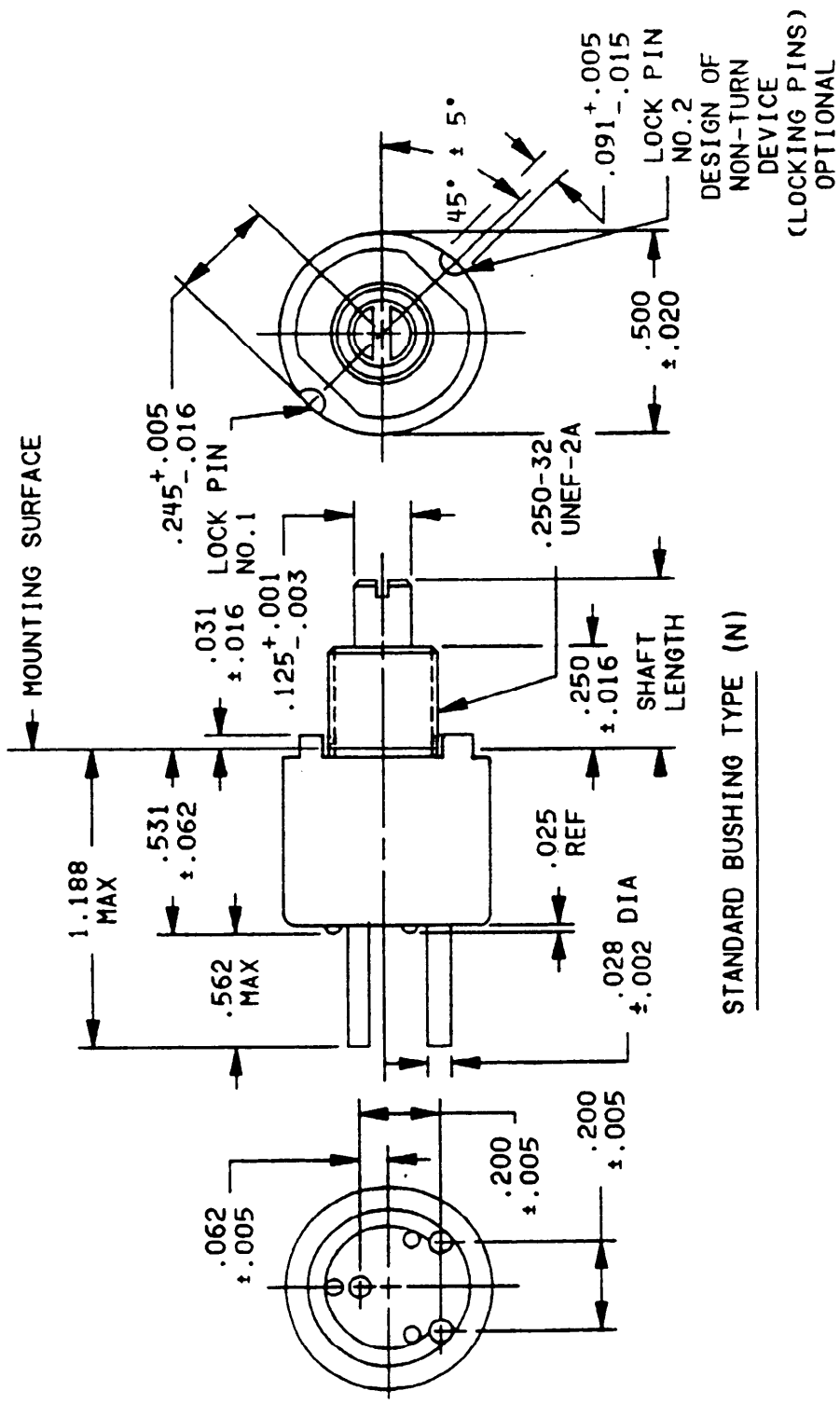


FIGURE 1. Style RV8 resistors.

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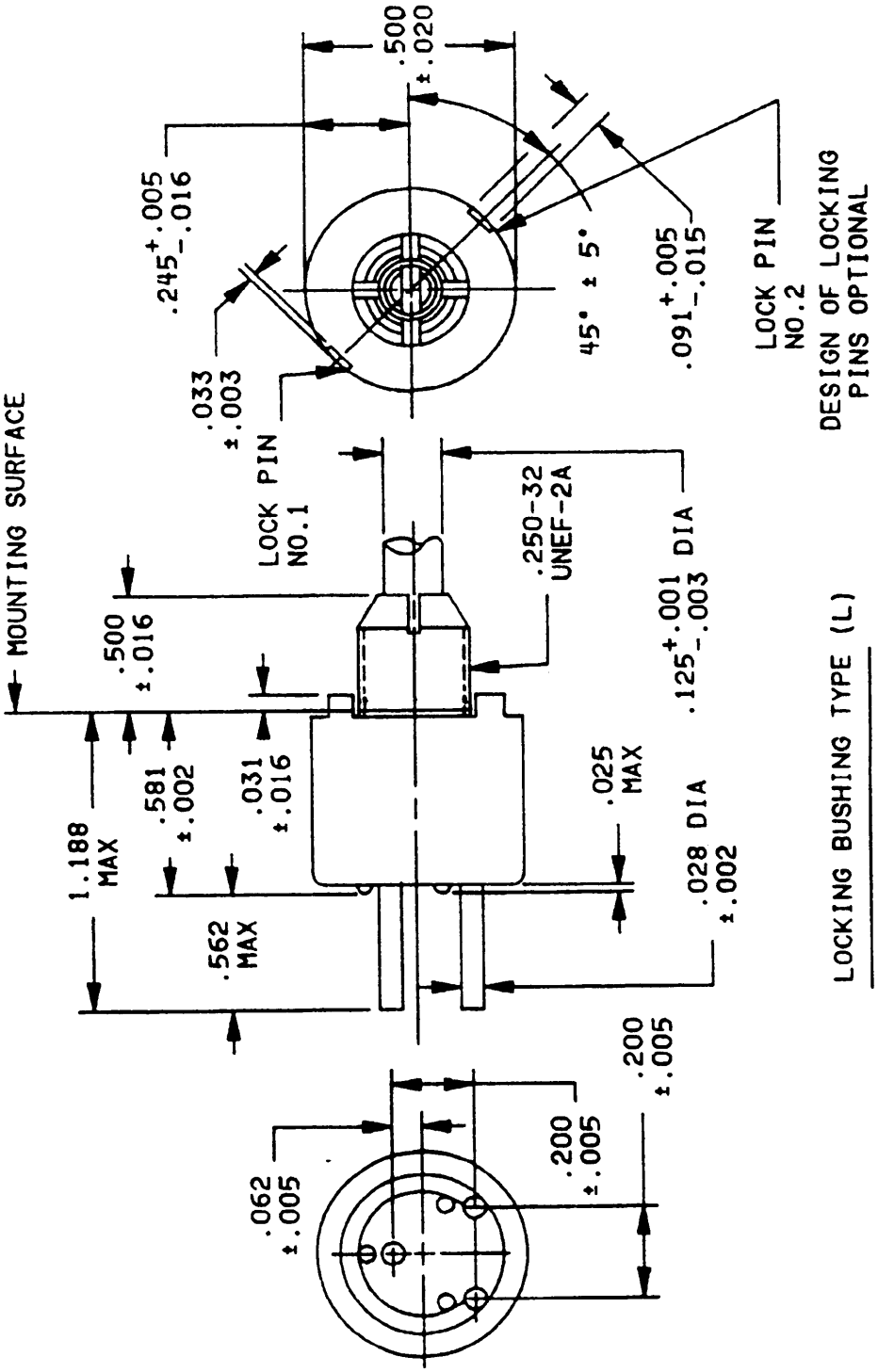


FIGURE 1. Style RV8 resistors - Continued.

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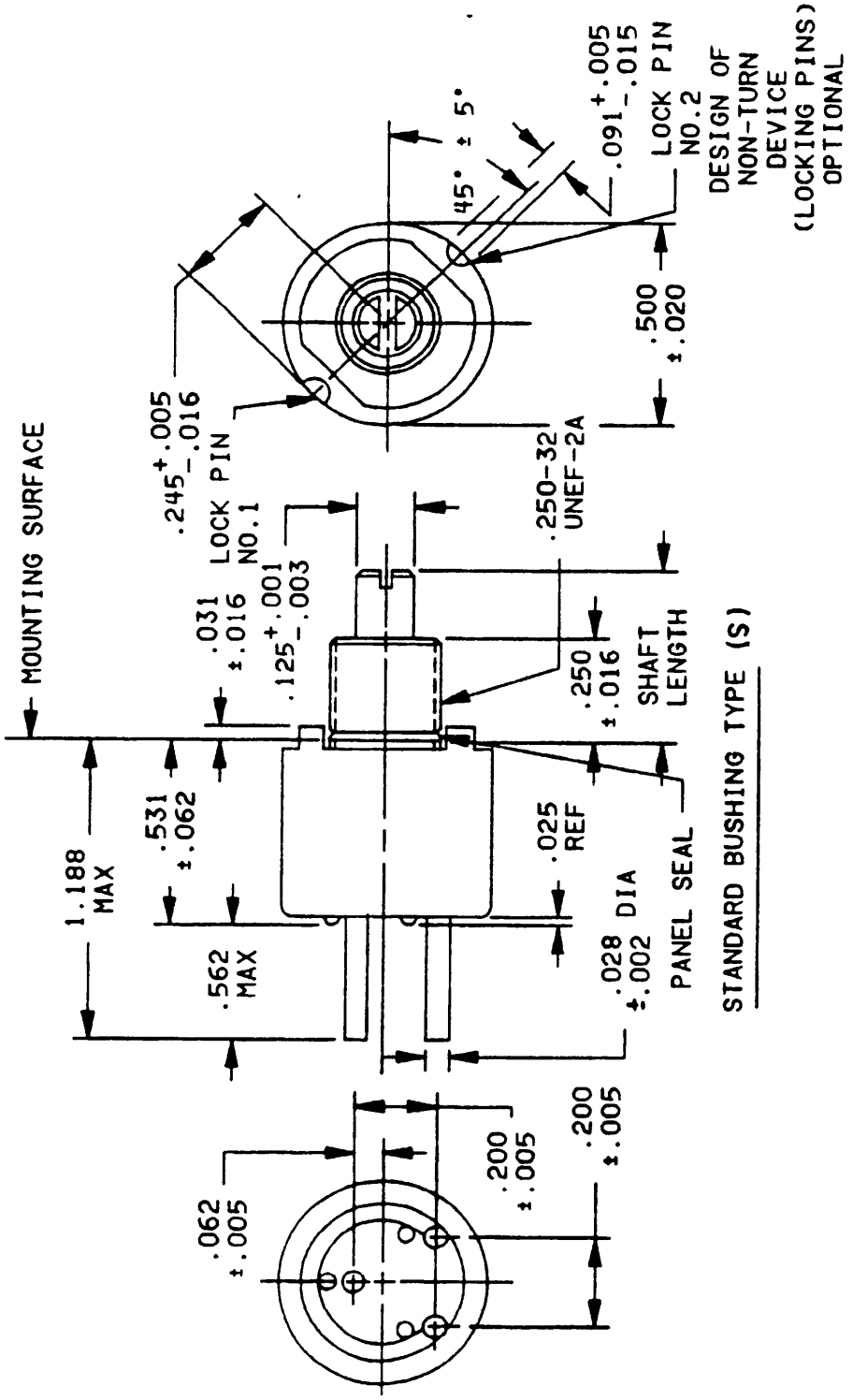


FIGURE 1. Style RV8 resistors - Continued.

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OPTIONAL

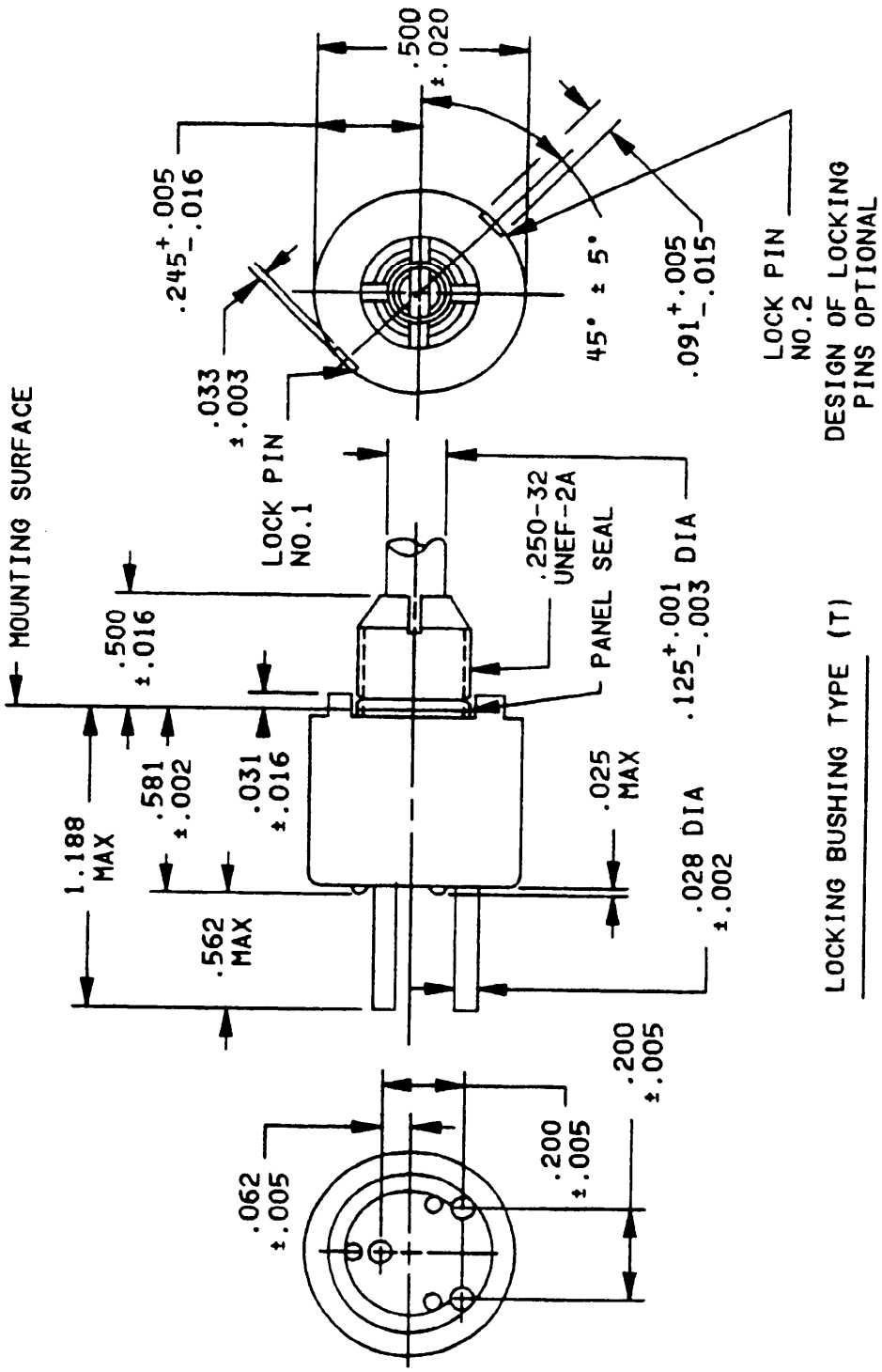
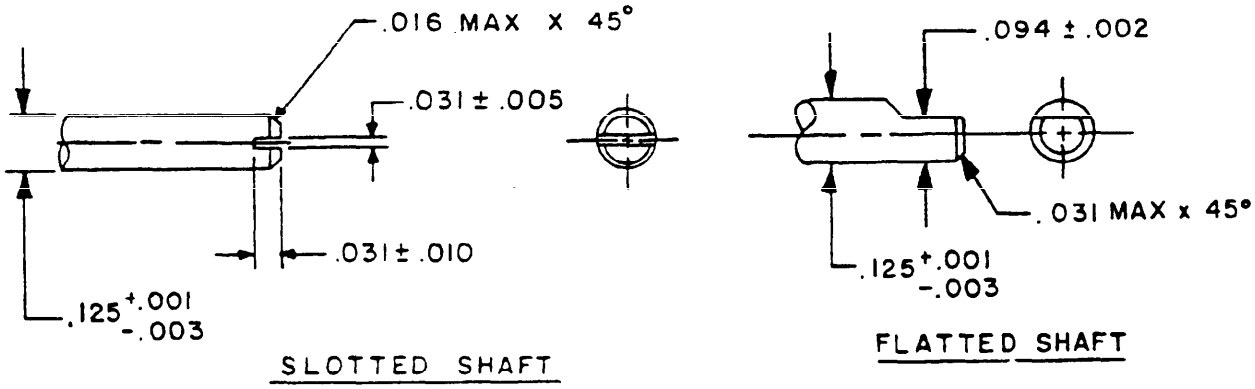


FIGURE 1. Style RV8 resistors - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.250	6.35
.002	0.05	.033	0.84	.453	11.51
.003	0.08	.062	1.57	.500	12.70
.005	0.13	.091	2.31	.531	13.49
.010	0.25	.094	2.39	.562	14.27
.015	0.38	.125	3.18	.581	14.76
.020	0.51	.200	5.08	.688	17.48
.025	0.64	.234	5.94	.812	20.62
.028	0.71	.245	6.22	1.88	47.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
4. When terminals are located symmetrically, the contact terminal shall be identified on the unit. The identifying mark shall be at the option of the supplier.

FIGURE 1. Style RV8 resistors - Continued.

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3.5 Nominal total resistance and rated continuous working voltage. The nominal total resistance and rated continuous working voltage shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F
Ohms			Megohms		
100	7	5	.10	224	160
200	10	7	.20	316	200
250	11	8	.25	350	200
500	16	11	.50	350	200
1,000	22	16	1.0	350	200
2,000	31	22	2.0	350	200
2,500	35	25	2.5	350	200
5,000	50	36	5.0	350	200
10,000	71	50			
20,000	100	70			
25,000	112	80			
50,000	158	112			

1/ Rated continuous working voltage at +70°C.

3.6 Minimum resistance. The maximum value of minimum resistance shall be 15 ohms for total resistance values of 100 to 500 ohms inclusive.

3.7 Power rating. The power rating shall be 0.5 watt per taper A resistors and .25 watt for tapers C and F resistors. Derate power 50 percent for nonmetallic panel mounting.

3.8 Torque.

3.8.1 Operating. The torque required to effect rotation shall be not less than 0.50 ounce-inch nor more than 6 ounce-inches.

3.8.2 Stop. Stops shall withstand a torque of 3 inch-pounds applied through the operating shaft.

3.8.3 Locking bushing. For the locking bushing type, the locking nut shall be tightened with a torque of 8 inch-pounds. After the locking nut is tightened, the contact arm shall not move when a torque of 20 ounce-inches is applied to the shaft.

3.9 Thermal cycling. The resistors shall not change in resistance in excess of 4 percent.

3.10 Immersion. When resistors are tested as specified in 4.6, no more than four bubbles shall be emitted.

3.11 Resistance to solvents. When resistors are tested as specified in 4.7, there shall be no evidence of mechanical damage and markings shall remain legible.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-94 and as specified herein.

4.2 Quality assurance. Quality assurance shall be as specified in MIL-R-94, except immersion and resistance to solvents testing.

4.3 Additional qualification inspections. Additional qualification inspections are immersion and resistance to solvents; these shall be added prior to solderability test as specified in table VIII, group IA of MIL-R-94.

4.4 Additional group A inspections. Additional group A inspections are immersion (see 3.10 and 4.6 with AQL of 1 percent) and resistance to solvents (see 3.11 and 4.7 with AQL of 4 percent); these shall be included in table X of MIL-R-94 prior to total resistance test.

4.5 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential shall be 750 volts rms at atmospheric pressure, and 350 volts rms at a reduced barometric pressure.

4.6 Immersion. Resistors shall be immersed for 1 minute ±5 seconds in water at 85°C +5°C, -0°C.

4.7 Resistance to solvents. Resistors shall be tested in accordance with method 215 of MIL-STD-202.

5. PACKAGING

5.1. Packaging. The requirements for packaging shall be in accordance with MIL-R-94.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-94 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Extension of qualification. Qualification to style RV6 will qualify style RV8 with the following tests: Immersion and resistance to solvents.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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